

THEO (CHIN WANG) WONG

ccww010801@gmail.com

linkedin.com/in/theo-wong-5562b3218 · github.com/Theo-Wong-67 · Right to live and work in Hong Kong SAR and the United Kingdom

RESEARCH BACKGROUND

I am an MSc Biomedical Engineering student at Imperial, with a background in computational modelling and signal processing. Most of my research was carried out in MATLAB and Python, and have worked on projects on neural decoders, grid cell models, and signal processing pipelines.

EDUCATION

MSc Biomedical Engineering (Neurotechnology Stream) — Imperial College London

Sept 2025 – Sept 2026 (expected) · Awarded the Department of Bioengineering MSc Scholarship

Provisional results: A in Human Neuromechanical Control and Learning; Neuroscience; Systems Physiology; Medical Device Certification; Mathematical Methods for Neural Science and Engineering; Data Analysis for Research. B in Principles of Biomedical Imaging; Brain Machine Interfaces; Computational Neuroscience; Journal Club. Individual research project in progress.

BEng (Hons) Biomedical Engineering — University of Reading

Sept 2022 – July 2024 · First-Class Honours · Credit-weighted average 73% across 240 credits

Selected marks: Biocybernetics 87; Brain–Computer Interfaces 83; Biomedical Problem Solving 79; Signal Processing 77; Introduction to Bioinformatics and Computational Biology 76; Biomechanics and Soft Technologies 75.

TECHNICAL SKILLS

Programming: Python, MATLAB, C++

Prototyping: AutoCAD, Solid Edge

Motor control and biomechanics: Kinematics, Jacobians and redundancy resolution for multi-joint limbs; arm dynamics and inter-joint coupling; muscle and joint stiffness; feedback and impedance control of reaching

State estimation and sensory integration: Kalman filtering as a model of how the brain estimates state; combining vision and proprioception; compensating for sensory delay; models of motor adaptation and learning

Machine learning: scikit-learn, neural networks, PyTorch, classification of noisy physiological signals, LDA, dimensionality reduction

Modelling and numerical methods: Attractor networks, spiking and rate-based neuron models, temporal-difference learning, Fourier and spectral analysis, ODE

Experimental Statistical methods: Experimental design, hypothesis testing, effect sizes and confidence intervals, model–data comparison

XR and interaction: Helped task design a VR training toolkit for site engineers (2021).

Languages: English, Cantonese and Mandarin (all native)

RESEARCH EXPERIENCE

Modelling Grid Cell Disruption in Alzheimer’s Disease

MSc individual research project, Imperial College London · 2025–26 · Supervisor: Prof. Simon Schultz

- Built a continuous attractor network model of grid cells in MATLAB to test how damaged synapses and noisy self-motion signals break down the brain’s internal map of space — the failure behind navigation problems in early Alzheimer’s disease.
- Added two damage parameters and a Fourier-based measure of grid regularity, reproducing the collapse reported by Ying et al. (2023, Current Biology).
- Set the agreement criterion from published effect sizes rather than choosing a tolerance by eye, so the model and the data could be compared statistically.
- Presented the work to a specialist audience, and worked with my supervisor and outside collaborators throughout.

Real-Time Neural Decoder for a Brain–Machine Interface

Group course project, Imperial College London · 2025–26 · Decoder designed and implemented independently

- Built a causal decoding pipeline in MATLAB that turns motor cortex spike trains into continuous 2D hand trajectories.
- Smoothed and binned the spike trains, classified reach direction with LDA, then estimated the trajectory with a Kalman filter per direction.

Synthetic ECG Generation and Filtering for Pathological Signal Classification

BEng research dissertation, University of Reading · 2023–24 · Supervisor: Prof. Sillas Hadjiloucas

- Built a synthetic ECG dataset with realistic noise and movement artefacts, giving a known ground truth to test processing against.
- Compared filtering methods and measured how each one changed classification accuracy on pathological signals.

SELECTED COURSEWORK IMPLEMENTATIONS (MATLAB)

Human Neuromechanical Control and Learning

Imperial College London · 2025–26

- Modelled the brain tracking a moving target from noisy vision and proprioception, and how well it can predict through sensory delay.
- Simulated a two-joint arm under feedback control, relating the controller to muscle stiffness and damping, and examined why fast movements lose accuracy.
- Worked through the kinematics of a multi-joint arm and the respective endpoint and joint stiffness.
- Modelled how people adapt to a force field, and the learning mechanisms.

Technical Documentation under EU MDR — Class I EMG/MMG Device

Medical Device Certification, Imperial College London · 2025–26

- Wrote the verification and validation section of a technical documentation file, for a portable EMG/MEG device under EU MDR Annex II, covering EMC testing and software validation to IEC 62304 Class A.

PROFESSIONAL EXPERIENCE

Quality Assurance Summer Intern — The Hong Kong and China Gas Company

June – August 2021, Hong Kong

- Assisted in the design of a VR training tasks for site engineers, developing interactive tasks for safety training coupling underground pipelines.
- Carried out internal safety audits and kept laboratory records in a regulated environment.

Online Assistant — Sainsbury's, London

Sept 2024 – June 2025

- Part-time work between degrees.

AWARDS

Department of Bioengineering MSc Scholarship — Imperial College London, 2025

REFEREES

Available on request.